

Runbook — One inverter has failed (the house still has some power)

Use this page when one of the two Sol-Ark inverters in the power room has gone dark, is has its **Alarm** indicator lit, or SolarAssistant shows one inverter has stopped reporting — and the other one is still running. It tells you how to find out which unit failed, try one safe restart, keep the house running on the survivor, and what to do if the failed one is the **master** (the one in charge). Promoting the other inverter to master is an installer job; it is written out here so a technician can talk you through it on the phone.

▲ STOP — read before touching anything

- **Never open an inverter case.** There is nothing inside you can fix, and the capacitors hold a lethal charge after power-off.
- **The only things you touch** on an inverter are: the breakers, the round **PV disconnect** switch on its side, the **power button**, and the **touchscreen**.
- **Power-OFF order is master first, then slave. Power-ON order is slave first, master last.** Doing it backwards makes the units fault and can leave the system down.
- **Batteries before inverters, always.** The battery packs must be on and showing lights before any inverter is switched on.
- **Never leave two inverters both set to Modbus SN 1.** Two "masters" will fight and neither will work.
- **Moving wires inside the inverter's comms terminal strip is a technician job.** Do not do it alone unless you have been shown how.

Words used on this page

- **Master / slave** — the two inverters work as a team. The **master** (Modbus SN 1) is in charge: it talks to the batteries, starts the generator, and copies its settings to the **slave** (Modbus SN 2).
- **Modbus SN** — the number (1 or 2) each inverter is given on its Parallel settings page.
- **MPPT** — one of the three solar inputs on each inverter. Each has 14 panels.
- **DIP switch** — a tiny two-position switch inside the comms terminal cover. It marks the "ends" of the link cable between inverters. In a two-inverter system **both are ON** and should not need touching.
- **Two-wire start** — a pair of small wires from the master inverter (pins 7 & 8) to the generator. When the master closes that circuit, the generator starts.

What you see → what to do

What you see	Go to
One inverter's screen is dark, or its Alarm indicator is lit, and the other is running normally	Procedure 1
Both inverters show F41 on their alarm list and one of them keeps faulting	Procedure 1, then 2
SolarAssistant shows one inverter column frozen / "stale", the house has power	Procedure 1
You did a restart and the same unit faulted again	Procedure 3
The dead inverter is inverter 1 (the master) — the generator did not start when the batteries got low, or the survivor shows F58 / F46	Procedure 4, now
The house went completely dark (both inverters off)	This is not a one-inverter failure. Go to the Batteries runbook, "inverter has shut down".
Replacement inverter has arrived	Procedure 5

Procedure 1 — Find out which inverter failed and what it says

- □ Go to the power room. Look at the four small lights on the front of each inverter: **DC, AC, Normal, Alarm**. You should see one unit with the **Normal** indicator lit and one with the **Alarm** indicator lit, or nothing lit at all.
- □ Write down which unit it is. Inverter 1 is the master; inverter 2 is the slave. (If you are unsure, tap * → **Basic Setup** → **Parallel** tab on a working unit: it shows *Master* or *Slave* and its Modbus SN.)
- □ On the failed unit, if the screen is alive, tap the home screen → **System Alarms**. Write down every code and the time. You should see one or more **F-numbers**. **If not** → if the screen is dark, note that; the power cycle in Procedure 2 is still the next step.
- □ Look at the alarm list on the **working** unit too. You will usually see **F41** there — that is the healthy unit reporting that its partner dropped out. **It is not a second failure.**
- □ Check the network cable between the two inverters (yellow, the only one of its kind on the comms strip) (it runs between the **Parallel** ports on the comms strip) is clicked in at both ends. **If the failed unit's code is F29 or F46** → a

loose cable is the most likely cause. Re-seat it and go to Procedure 2.

- □ If the alarm list shows the same code **five or more times** in a row, the inverter has **locked itself out** and will not restart on its own. Procedure 2 is still the right next step — the power cycle is the manual reset.

Common codes, in plain words:

Code	What it means
F41	The other inverter stopped. Seen on the healthy unit. Normal.
F29 / F46	Can't talk to the other inverter — cable or address problem.
F61	Someone pressed the slave's power button without turning the master off first.
F58	Can't see the batteries. Expect this on the slave if the master has died (the battery cable lands on the master).

Procedure 2 — One full restart, in the right order

Most faults clear with one restart. Do this **once**. If the same unit faults again afterwards, go to Procedure 3 — repeated restarts do not help and can make things worse.

You will need: a flashlight, a phone or notepad, about 15 minutes. The house will be without power for a few minutes in the middle of this.

- □ Warn everyone in the house the power is going off for a few minutes.
- □ **On the master (inverter 1) first:** switch OFF its three AC breakers — **GRID, GEN and LOAD**. Then turn the round **PV disconnect** switch on its side to OFF. Then press its **power button** to OFF. The screen goes dark after a few seconds.
- □ **Now the slave (inverter 2):** the same three steps — AC breakers OFF, PV disconnect OFF, power button OFF.
- □ Switch OFF the battery breakers (order does not matter).
- □ **Wait a full minute.** This lets everything discharge.
- □ Switch the battery breakers back ON. Walk to the battery rack and confirm **every pack shows lights** (the indicator labelled **RUN** blinking slowly, or the **SOC** capacity bar lit). **If not** → see the *Batteries* runbook, "power everything up in the right order", before going further. Inverters must not come on with the batteries dark.
- □ **Slave (inverter 2) first:** power button ON → wait for the screen → PV disconnect ON → wait for the **DC** indicator to light → AC breakers ON (GRID, GEN, LOAD).
- □ **Master (inverter 1) last:** the same steps.
- □ Watch both screens for **up to 3 minutes**. You will probably see **F29** and **F41** flash up briefly while they find each other — that is expected. Then the **Normal** indicator should light on both units.
- □ **If both come up Normal** → done. Check SolarAssistant shows both inverter columns updating. Keep an eye on it for a day.
- □ **If the same unit faults again** → Procedure 3.

▲ WARNING: If the unit that failed is the **master** and it does not come back, go to **Procedure 4 now** — the generator will not start on its own until this is sorted, and the batteries will drain.

Procedure 3 — Isolate the dead inverter and run the house on the survivor

Goal: the failed unit is switched fully off and left alone; the good unit carries the house until the installer comes.

You will need: a flashlight, a piece of tape and a marker to label the dead unit.

- □ On the **failed unit only:** AC breakers OFF (GRID, GEN, LOAD) → PV disconnect OFF → power button OFF → its battery breaker OFF. Tape a note on it: "FAILED — DO NOT SWITCH ON — date".
- □ Leave the inverter-to-inverter link cable (yellow) where it is. In a two-inverter system the survivor is already an "end" of the link and its DIP switch stays ON. Nothing to change.
- □ Restart the survivor on its own (Procedure 2 steps, but only for that unit) so it stops looking for its partner. It may show F29/F41 once; it should then go **Normal**. **If not** → call Sol-Ark support with the code.
- □ **Shed load.** One inverter alone can supply about **12 kW from the batteries** (15 kW when the sun is strong) and **half the usual solar** — the dead unit's 42 panels are out of action. In practice: **run one big thing at a time**. Do not run the well pump, clothes dryer, electric range/oven, HVAC, water heater and EV charger together. Two of those at once is usually fine; four is not.
- □ Nothing to change in SolarAssistant. The dead unit's column stays frozen until the replacement is wired up; that is expected.
- □ Call the installer (see *Who to call*) and give them the codes you wrote down.

Generator charging still works if the **master** is the survivor. The charge current into the batteries is set per inverter, so with one unit gone the batteries charge at half the usual rate — that is normal, and the generator run will simply take longer.

Procedure 4 — If it is the MASTER (inverter 1) that died

This is urgent. The master is the only unit that (a) talks to the battery packs and (b) can start the generator. With it dead, **the generator will not auto-start and the batteries can run all the way down** with nothing to stop it. Do not wait until morning.

Part A — Stopgap: get the generator running by hand (you can do this)

You will need: a flashlight, the generator's enclosure key if it is locked.

- Write down the current battery % from SolarAssistant or the survivor's screen. Below about **20 %** you have very little time.
- Go to the generator. Open the **left rear door**. On the **DSE** controller panel, press the **Manual** mode key (second from left on the bottom row; the small indicator beside it comes on), then press the **Start** key (far right of the bottom row) once. It cranks and should run within a few seconds. **If not** → see the *Generator* runbook, "generator won't start".
- Let it run unloaded for about 3 minutes.
- Back in the power room, watch the survivor's **AC** indicator. The generator is wired into both inverters, so the survivor is being fed as soon as the generator runs. **If the AC indicator comes on** within 2–3 minutes, the survivor is taking generator power — the house is on generator and the batteries are charging. You have time. **If the AC indicator does not come on** → the survivor is refusing the generator without a master present. Turn off every big load (see Procedure 3) and go straight to Part B with the installer on the phone.
- Leave the generator running in **Manual** until the installer tells you otherwise. It will not stop by itself in Manual — **check the fuel**.

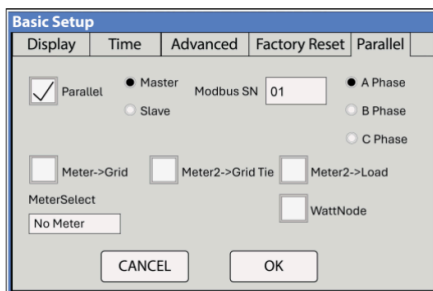
▲ WARNING: A generator in **Manual** will run until you stop it. Do not leave the property with it running unattended for long periods, and do not return it to Auto until a working master inverter is in charge again.

Part B — Make the slave the new master (installer job — do not do this alone)

▲ WARNING: Part B involves moving small wires inside the inverter's comms terminal strip and the battery communication cable. Do this only with the installer or Sol-Ark support on the phone, or if you have been shown it in person. Done wrong, the batteries and generator stay invisible to the system and the house can lose power.

You will need: a small flat screwdriver for the pluggable terminal strip, a flashlight, the generator off, the phone with the technician on it.

- Power everything down in the right order (Procedure 2, the OFF half): dead master first if it is alive at all, then the slave, then the battery breakers. Wait a minute.
- **Move the battery communication cable.** It is a network-style cable from the master battery pack to the dead master's right-hand RJ45 jack labelled **Battery CANBus**. Unplug it there and plug it into the **same jack on inverter 2**. (If each inverter has a small splitter on that jack, move only the battery leg.)
- **Move the generator start wires.** Two thin wires land on pins **7 and 8** of the dead master's comms terminal strip. Loosen the screws, move them to pins **7 and 8 on inverter 2**, tighten. Polarity does not matter — it is a simple switch contact.
- **Leave the SolarAssistant cables alone.** They are not affected.
- Power up the batteries (all packs showing lights), then **inverter 2 only**.
- On inverter 2's screen: ⌘ → **Basic Setup** → **Parallel** tab. Tap **Master**, set **Modbus SN** to **01**, leave *Parallel* ticked, tap **OK**.



Parallel setup tab

The *Parallel* tab. Tap "Master" and change Modbus SN to 01. Nothing else on this screen changes.

- Power inverter 2 off and on again (power button) so the new role takes effect. Wait for the **Normal** indicator to light.

- □ With the technician, **check every settings page** on the new master: Grid type (120/240 V split-phase), **Battery Setup** (capacity 61.4 kWh, Shutdown 15 %, charge amps, "BMS Lithium Batt 00", "Activate Battery" ticked), generator **Start %** (35 %), **Max Gen Runtime** (150 min), and — if Time of Use is on — that **Charge** is ticked in every time slot. These were copied from the old master, but confirm them.
- □ **Prove the generator:** * → **Battery Setup** → **Charge** tab → tick **Gen Force** → OK. Within 2 minutes the home screen shows **Gen Signal** and the generator starts (the DSE must be back in **Auto** for this). Then untick **Gen Force**; the generator stops after its cool-down. **If not** → the start wires are on the wrong pins or the DSE is still in Manual.
- □ **Prove the batteries:** * → **System Setup** → **Li-Batt Info** shows live battery voltage, % and current. **If the screen is blank** → the battery cable is not seated in the Battery CANBus jack.
- □ Shed load as in Procedure 3 until the replacement is installed.

Procedure 5 — When the replacement inverter arrives

This is the installer's job. What to expect, so you can check it was done:

- □ Firmware on the new unit is updated to **match** the existing one before it is connected (both must show the same COMM and MCU numbers on System Setup).
- □ The new unit is set up on its Parallel tab as the **slave**, Modbus SN **02** (simplest: leave the promoted unit as master), or the wiring from Procedure 4 Part B is moved back and it becomes master again. **Never both at 01.**
- □ Its DIP switch is set ON (two-inverter system: both ON).
- □ Full power-up in order: batteries → slave → master. Brief F29/F41, then both Normal.
- □ Installer confirms the Grid and Battery settings copied across to the new unit.
- □ SolarAssistant's RS485 lead is plugged back into the new unit's Battery CANBus splitter; its column starts updating within a few minutes.

Stop and call for help when...

- The same unit faults again after **one** restart (Procedure 2). Do not keep restarting.
- Any code you see is **not** in the table above, or the screen shows anything about **arc fault (F63)**, **over-voltage (F55)**, or **DC insulation (F24)**.
- The **master** has died — call while you do Procedure 4 Part A.
- The survivor will not go **Normal** after being restarted on its own.
- You smell burning, see scorch marks, hear buzzing or crackling from an inverter.
- The battery % is under **20 %** and falling and the generator is not running.

Who to call

Who	When	Contact
Installer — Ernie Williams, Mainstream Green Solutions, Lexington TN	First call for anything on this page; replacement and promotion work	(731) 697-1665 · Ernie.williams@mainstreamgreensolutions.com · www.mainstreamgreensolutions.com
Sol-Ark Technical Support	Fault codes, settings, walking you through the screens	7 days a week (not 24 h) — support@sol-ark.com

Reference — for the technician

Source: Sol-Ark *15K Installation Manual* MA-00007 **Rev. 13** (July 2026), §2.12 Power Cycle Sequence, §5 Parallel Systems, §8.1 Error Codes. Applies to 120/240 V split-phase parallel systems on a shared battery bank. Site: 2 × 15K-2P, 12 × Pytes V5, off-grid, Patriot 40 kW on the GRID inputs.

What the master does that a slave doesn't

Responsibility	Manual reference
Holds Modbus SN 1 ; its GRID and Battery settings are copied to every slave	§5.2 step 3; §4 power-on step C
Controls the generator two-wire start (relay on pins 7 & 8)	§5.2 note: <i>"The inverter assigned as Master will control the two-wire start feature"</i>
Carries the CT pair (grid-tied sites only)	§2.9: <i>"Only one pair of CT sensors must be wired to the designated Master inverter"</i>
Is the only unit whose Battery CANBus jack is cabled to the BMS on a typical build — slaves take battery state over the parallel link	Site wiring convention; see page 00

Everything else — LOAD, GRID, GEN AC connections, the battery bank — is shared by every unit (§5.1 C: *"All INPUTS/OUTPUTS must be shared among ALL parallel inverters, except for DC solar inputs"*). PV strings are **not** shared: a dead inverter takes its

own strings with it.

Fault codes (§8.1)

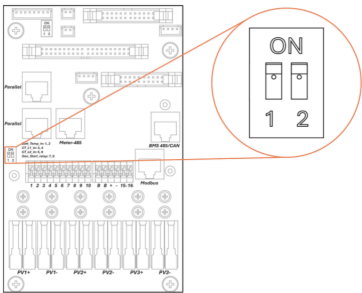
Code	Name	Meaning in a parallel system
F29	Parallel_Canbus_Fault	Lost parallel comms. "Check cables, and Modbus addresses." Normal for a moment during power-up until all units are on.
F41	Parallel_System_Stop_Fault	"If one system faults in parallel, this normal fault will register on the other units as they disconnect from the grid." The healthy units are reporting a neighbour's failure.
F46	Battery_Backup_Fault	"Cannot communicate with other parallel systems. Verify that the Master is set to 1, the Slaves are set to 2-9, and the Ethernet cables are connected."
F61	Button_Manual_OFF	"The parallel Slave system turned off without turning off the Master." Someone pressed a slave's power button.
F58	BMS_Communication Fault	Unit is in lithium BMS mode but has no BMS link — expect this on slaves if the master (which holds the CAN cable) goes down.

Ground rules

- **Power-cycle order (§2.12, parallel):** OFF — AC breakers (GRID, GEN, LOAD), then PV DC disconnect, then power button, **master first, then slaves**; battery breakers last, any order; wait ~1 min. ON — reverse it: **slaves first, master last**. "Inverters will likely fault momentarily with F29 and F41 codes until all inverters are ON." (§5.2 step 6)
- **DIP switches (§5.1):** termination on the two ends of the parallel daisy-chain. Two units: **both ON**. Three or more: first and last ON, middle units OFF. Any change to the chain means re-checking these. In a 3+ system, a dead middle unit must be cabled around (Parallel_1/Parallel_2 ports) and the two new ends re-terminated.

DIP Switch Configuration for Parallel Systems

In parallel systems, set the "DIP Switches" as shown, according to the table below.



Inv 1 (Master)	Inv 2	Inv 3	Inv 4	Inv 5	Inv 6	Inv 7	Inv 8	Inv 9	Inv 10	Inv 11	Inv 12
OFF											
ON	ON										
OFF	ON	OFF									
OFF	ON	ON	OFF								
OFF	ON	ON	ON	OFF							
OFF	ON	ON	ON	ON	OFF						
OFF	ON	ON	ON	ON	ON	OFF					
OFF	ON	ON	ON	ON	ON	ON	OFF				
OFF	ON	ON	ON	ON	ON	ON	ON	OFF			
OFF	ON	ON	ON	ON	ON	ON	ON	ON	OFF		
OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON	OFF	
OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON	OFF

Parallel systems with 2 inverters must have their DIP switches in the ON position

Installation Manual Rev 13, §5.1. Two-inverter systems: both switches ON.

- **Firmware:** all units in parallel must show the same COMM and MCU version (§5.1 A). A replacement unit must be updated to match *before* it joins the chain.
- **Five strikes.** Self-clearing alarms auto-restart up to 5 times; the fifth stop on the same alarm locks the unit out until it is manually reset and the cause is cleared (installer's deck). A unit that is "dead" may just be locked out — read its alarm list before condemning it.
- **Batteries up before inverters.** Pytes V5 manual §5.4: power all packs, press the master pack's SW for 1 s, wait for every pack to show lights, *then* power the inverters — the inverter capacitors' inrush can shock an unready pack. Shut down in the opposite order (inverters first, then hold the master pack's SW 3 s).
- **Never run parallel mode without a battery bank** (§5.1) — irrelevant during a failure but the reason you don't "just" pull the battery to isolate a unit.
- **Charge current:** Gen/Grid "A" on the Battery Setup page is **per inverter** — the total into the bank drops automatically with one unit gone. Raise the per-unit value only if the generator can carry it.
- **Master-failure stopgap:** the generator is on the GRID input of every unit (§5.1 C), so surviving slaves are physically fed the moment it runs. Whether a slave's logic qualifies that input with no master on the chain is **not stated in the manual** — confirm with Sol-Ark support before it is needed.

- **SolarAssistant:** no reconfiguration after a failure or promotion. `/inverter/settings` shows the promoted unit as master; the dead unit's column stays stale. If low-SOC automations key off a specific inverter's values, re-point them.

Capacity

Parallel Systems Sol-Ark 15K-2P-LV @ 120/240 V Split-Phase

# of inverters in parallel	Continuous output power with PV (kW)	Continuous output power with batteries (kW)	Max. Grid Passthrough Current (A)	Peak power 10 sec (kVA)
1*	15	12	200	24
2	30	24	400	48
3	45	36	600	72
4	60	48	800	96
5	75	60	1000	120
6	90	72	1200	144
7	105	84	1400	168
8	120	96	1600	192
9	135	108	1800	216
10	150	120	2000	240
11	165	132	2200	264
12	180	144	2400	288

§5.1 table: per unit 15 kW continuous with PV, 12 kW on battery, 200 A grid passthrough, 24 kVA for 10 s.

Promotion checklist (technician)

1. Full power-down per §2.12. 2. Battery CAN → new master's Battery CANBus jack; two-wire start → pins 7 & 8; CT pair (grid-tied only). RS485 to SolarAssistant unchanged.
2. Parallel tab: Master, Modbus SN 01; other slaves stay 2...n. 4. Chain/termination as above. 5. Power up slaves first, master last. 6. Verify Grid type, Battery Setup (capacity, Shutdown/Restart, charge A, BMS Lithium Batt 00, Activate Battery), Gen Start %, Gen down time, Max Gen Runtime, TOU Charge ticks; confirm slaves report the same (§4 step E). 7. Gen Force → Gen Signal ≤2 min → start → clear. 8. Li-Batt Info live, no F58. Replacement unit: match firmware, Parallel ✓, vacated SN, DIP per position, cable in, full power cycle, verify settings copied. Never two units at SN 1.

Two-inverter quick card

	Slave (SN 2) dies	Master (SN 1) dies
Capacity left	15 kW PV / 12 kW battery	15 kW PV / 12 kW battery
Gen auto-start	still works	lost until promotion
BMS comms	still works	lost until CAN lead moved
Chain work	none; survivor's DIP stays ON	set SN 2 → 1; move CAN lead + gen start wires; DIP stays ON
Power-up order	just the survivor	just the survivor (it is now the master)
Expect	F41 logged on survivor	F41, F29/F46, F58 logged on survivor